

Passive Radiation Shielding and Fault-Tolerant Design

Protecting High-Altitude Electronic Systems from UV and Cosmic Radiation

High Altitude Areas (HAA) and Super High Altitude Areas (SHAA) of Ladakh region presents one of the harshest operating environments for electronic equipment and systems. Higher altitude means thinner atmospheric shielding, so electrical and electronic components receive more UV radiation and cosmic rays. This can cause occasional electronic errors such as:

1. Single-event upsets (SEUs): a bit in memory or registers changes unexpectedly.
2. Microcontroller crashes/resets
3. Corrupted sensor readings
4. Long-term radiation degradation of some electronic components

In order to tackle these challenges posed by radiation exposure, this proposed design aims at protecting the electronics from UV and radiation exposure at high altitude using radiative shielding, techniques like sensor redundancy using voting, watchdog+automatic recovery, sensor-induced error checking using range check and comparison among sensors.

1.Design Principles

- 1. UV shielding:** A hybrid PMMA-based thin-film coating incorporating surface-passivated TiO_2 and ZnO nanoparticles absorbs and scatters a large portion of incident UV, while reflecting visible and near-infrared solar radiation. This reduces UV exposure of the enclosure and lowers solar heat load.
- 2. Secondary UV protection and weatherable structural shell (PC/ASA blend):** An opaque, light-colored, pigmented polycarbonate/acrylonitrile styrene acrylate (PC/ASA) blend shell sits beneath the coating as a structural layer and secondary UV barrier. The ASA acrylate rubber phase gives improved UV and weathering resistance and reduced yellowing compared with plain polycarbonate, while the PC component provides impact strength and rigidity, retaining good toughness at low temperature. The pigmented shell blocks residual UV that passes the outer coating and protects seals and cabling from photodegradation.
- 3. Neutron/cosmic-secondary shielding:** Borated high-density polyethylene (HDPE) is used as an inner liner around the MCU and memory compartment. The hydrogen in the polyethylene slows (moderates) the atmospheric neutrons that dominate single-event upsets at altitude, and the boron captures them with little secondary gamma emission. The liner reduces rather than eliminates the neutron flux, so it complements the fault-tolerance measures described below.

4. Watchdog supervision and automatic recovery: Atmospheric neutrons can cause single-event upsets that hang the microcontroller or corrupt stored data. Rather than trying to predict these events, the system detects and recovers from them. A watchdog timer supervises the ESP32, calibration and configuration data are verified against stored checksums on a fixed schedule, and the controller restarts into a known safe state when a fault is detected.

2.Composite Layer Stack

A workable layer stack for the enclosure wall, ordered from the outer surface inward to the electronics compartment:

Layer	Material	Recommended Thickness	Role
Outer layer	Hybrid PMMA-based thin film with surface-passivated rutile TiO ₂ and ZnO nanoparticles	40-60 µm dry film	Absorbs and scatters UV; reflects visible/NIR solar radiation
Intermediate shell	Opaque, pigmented, weatherable PC/ASA blend	5-10 µm	Provides structural protection and acts as a secondary UV barrier by reducing transmission of residual UV radiation
Interfacial primer	UV-stable acrylic-silane tie layer	3-4 mm (minimum 2.5 mm)	Promotes adhesion between the PMMA coating and the PC/ASA shell; a flexible formulation helps accommodate thermal-expansion mismatch during diurnal temperature swings
Inner neutron liner	Borated HDPE (5–10% boron by weight), 10–20 mm thick, fitted around the MCU and memory compartment	20 mm (range 10-20 mm)	Moderates and absorbs atmospheric secondary neutrons, reducing the neutron flux reaching the MCU, memory and voltage reference

3.How This Behaves Physically

1. Solar UV, visible and near-IR radiation reaches the outer PMMA-based coating. The TiO₂ and ZnO nanoparticles absorb and scatter a large portion of the UV, while visible

and near-IR radiation is largely reflected, reducing UV exposure and solar heating of the enclosure.

2. Residual UV that passes the coating is blocked by the opaque, pigmented PC/ASA shell, so the UV reaching the electronics compartment is substantially reduced.
3. Atmospheric neutrons, the main cause of single-event upsets at altitude, pass through the enclosure wall and reach the inner liner. In the 10–20 mm borated HDPE liner fitted around the MCU and memory compartment, hydrogen slows the lower-energy neutrons and boron captures them with little gamma emission. Very high-energy neutrons and muons largely pass through, so the liner reduces rather than eliminates upsets and is used together with the watchdog, redundancy and integrity-check measures below.
4. Radiation-induced upsets are random and cannot be predicted, so the design relies on fixed-schedule supervision. Configuration and calibration checksums are verified periodically and after every reset, and the watchdog supervises the controller continuously.

To improve system reliability, a sensor-validity check is introduced before the measurements are used by the controller.

Proposed sequence:

Sensors → Sensor Validity Check → Thermal Controller → Watchdog Monitoring → Fault Detection → Automatic Recovery

The watchdog is serviced only from the main control loop, after the sensor validity check has passed, so a hung or corrupted program cannot keep it satisfied.

If the MCU becomes unresponsive due to a software fault or system disturbance, the watchdog timer detects the timeout and automatically resets the MCU.

After reset:

- Initialize the sensors and measurement interfaces.
 - Validate the sensor readings and reject invalid measurements.
 - Check the system operating parameters, including temperature and power/current measurements.
 - Check the thermal safety condition.
 - Put the heater/heat source into a safe OFF state.
 - Keep the cooling fan active if required for thermal recovery.
 - Resume normal monitoring and adaptive cooling control once valid sensor readings are available.
5. Radiation-Induced Sensor Error Rejection: Radiation-induced transient errors may produce corrupted or unrealistic sensor readings; therefore, sensor data is validated before being used for thermal control.

Rather than assuming every sensor reading is correct, we can introduce a **three-level validation system**:

Level 1: Range Check

Reject physically impossible values.

Example: Temperature = $+150^{\circ}\text{C}$ $\rightarrow$ Invalid

Level 2: Rate-of-Change Check

Reject sudden unrealistic jumps.

Example: $-15^{\circ}\text{C} \rightarrow -14^{\circ}\text{C} \rightarrow -13^{\circ}\text{C} \rightarrow$ **Normal**

but $-15^{\circ}\text{C} \rightarrow -14^{\circ}\text{C} \rightarrow +80^{\circ}\text{C} \rightarrow$ **Possible corrupted reading**

Level 3: Cross-Sensor Sensor Redundancy Using Voting Technique:

- Use **multiple temperature sensors** to obtain independent measurements of the thermal environment.
- The ESP32 compares the sensor readings and applies a **majority-voting technique** to determine the most reliable temperature value.
- If **two sensors provide similar readings** while one gives a significantly different value, the majority reading is considered valid.
- The abnormal sensor reading is identified as a **possible sensor fault** and excluded from the control decision.
- Example:
 $T1 = -10^{\circ}\text{C}, T2 = -11^{\circ}\text{C}, T3 = +80^{\circ}\text{C} \rightarrow$ **Voting result ≈ -10 to -11°C**
- The validated temperature is then used for **heater and fan control**.
- If the sensor readings do not provide a clear majority, the system can enter a **safe state** and generate an alarm.
- This redundancy helps prevent a single radiation-corrupted sensor reading from causing an incorrect heater or fan control decision.
- This approach improves **fault detection, reliability and tolerance to corrupted sensor readings**. We are using voting/redundancy at the critical decision level rather than duplicating the entire system.

Because remote Ladakh installations are often off-grid with limited maintenance access, the system is positioned as maintenance-reducing passive protection and adaptive fault tolerance.